

Routing Protocol: EIGRP & OSPF

Course Code: COE 3206

Course Title: Computer Networks



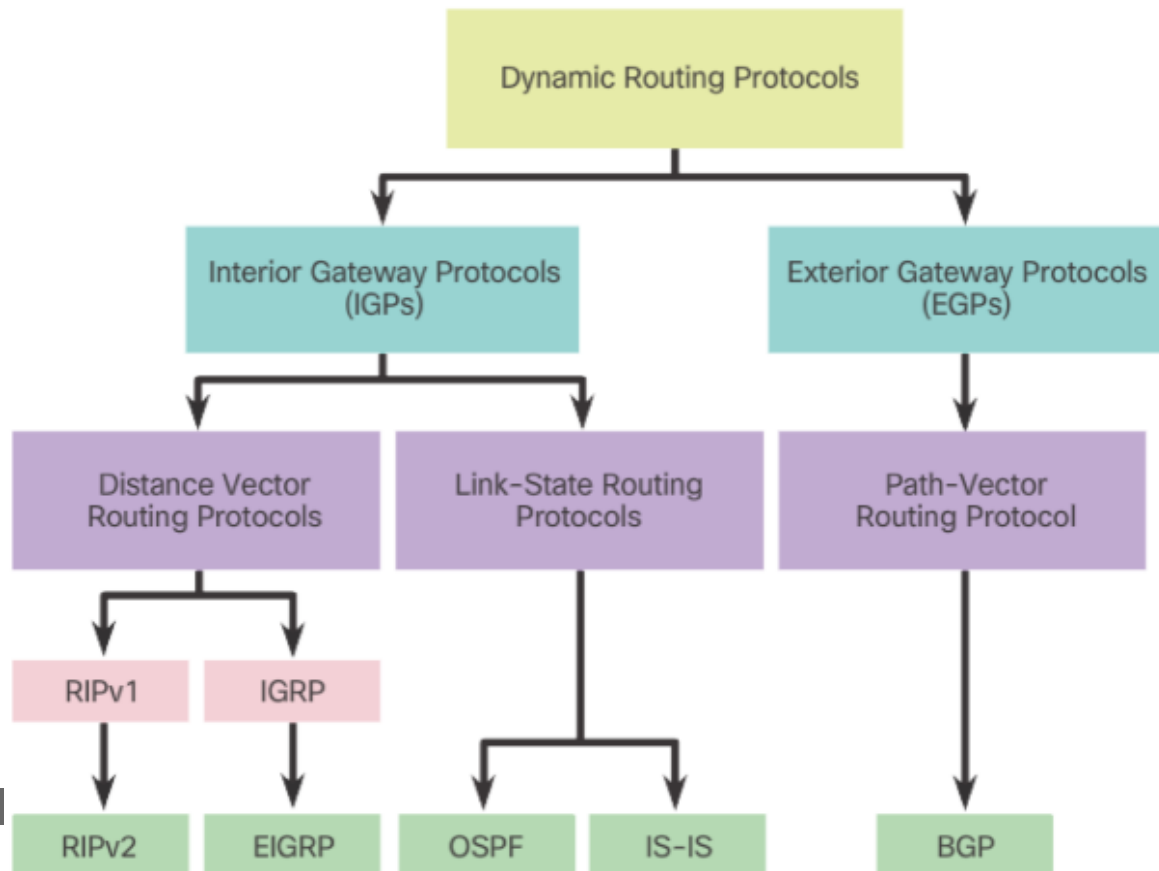
Dept. of Computer Science
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Lecturer No:	10	Week No:		Semester:	Fall 25-26
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Lecture Outline



1. Introduction
2. Metric
3. Neighbor Discovery
4. EIGRP
5. OSPF





Routing Protocol: EIGRP

RIP vs EIGRP



	RIP	EIGRP
1	It support maximum 15 routers in the network. 16 router is unreachable.	It supports maximum 255 routers in the network. However, the default is 100 routers. (highly scalable)
2	Slow convergence	Fast convergence due to feasible successor
3	In RIP routing protocol, we cannot create a separate administrative boundary in the network.	In EIGRP routing protocol we can create a separate administrative boundary in the network with the help of autonomous system No. Less routing Table exchange is required.
4	It calculates the metric in terms of Hop Count from source network to destination network.	It calculates the metric In terms of bandwidth and delay (default).
5	RIP works on Bellman Ford algorithm.	EIGRP works on DUAL(Diffusing Update Algorithm) Algorithm.
6	It only maintains the best route to each destination.	It maintains the best route and some other alternative routes for each destination.
7	It is basically use for smaller size organization.	It is basically use for medium to lager size organization in the network.

Metric



❖ Combination of different factors

- Bandwidth
 - Delay
 - Load
 - Reliability
- } Default



Bandwidth



- No. of bits that can be sent over a link (kbps)
- Depends on **interface type**
- Use *bandwidth* $<1 \text{ to } 10,000,000>$ command to set bandwidth in kbps
- This is not real bandwidth; **real bandwidth depends on clock rate**
- The *bandwidth* command only influence route selection by routing protocol
- If no bandwidth is set, the default bandwidth of an **interface** is considered
- Calculated as the lowest bandwidth among all links in a route

Bandwidth



Table A: Default bandwidth and delay

<i>Interface</i>	<i>Bandwidth</i>	<i>Delay (microseconds)</i>
Serial (T1)	1544 Kbps	20,000
Ethernet	10 Mbps	1000
Fast Ethernet	100 Mbps	100
Gigbit Ethernet	1000 Mbps	10
10 Gigbit Ethernet	10 Gbps	10

Delay



- *Delay* is a measure of the time for a **packet** to **reach its destination** over a route (**In theory**)
- In practice, **it is a constant set by the network engineer**
- To set delay for an interface, use *delay <value>* command
- The value can be anything between *10 to 167,772,140 microseconds*
- If it is not set, the default value (**Table A**) of each interface comes into effect
- Calculated as sum of delays in exit interfaces of all routers in a route.

Metric Calculation



$$Metric = \left[\frac{10^7}{least\ bandwidth} + delay_{total} \right] \times 256$$

Units

Bandwidth: kbps

Delay: Tens of microsecond

If the total delay is 30 seconds, $delay_{total} = 30/10 = 3$

Metric Calculation



$$\text{Metric} = \left[\frac{10^7}{\text{least bandwidth}} + \text{delay}_{\text{total}} \right] \times 256$$

Route: 1-4-2-B

Least BW = 56 kbps

Total delay = $100/10 + 100/10 + 2000/10 = 220$

$$\text{Metric} = \left[\frac{10^7}{56} + 220 \right] \times 256 = 45770496^*$$

Route: 1-3-2-B

Least BW = 128 kbps

Total delay = $100/10 + 100/10 + 1000/10 = 120$

$$\text{Metric} = \left[\frac{10^7}{128} + 120 \right] \times 256 = 20030720$$

Perform rounding in every steps of calculation

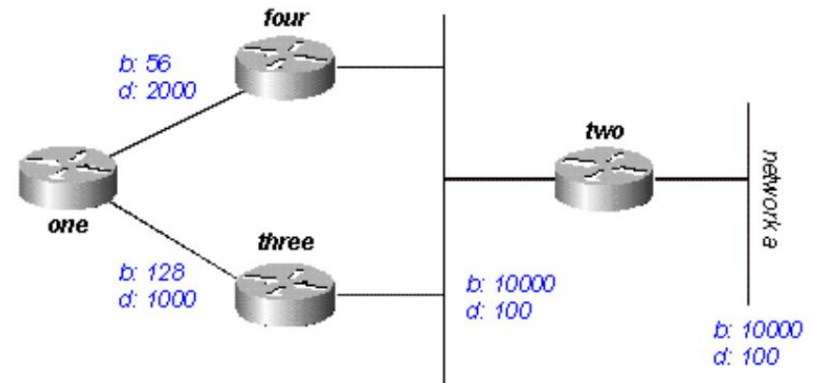


Fig. 1 A sample network

Exercise



Calculate metric for all possible routes from router ONE to network A

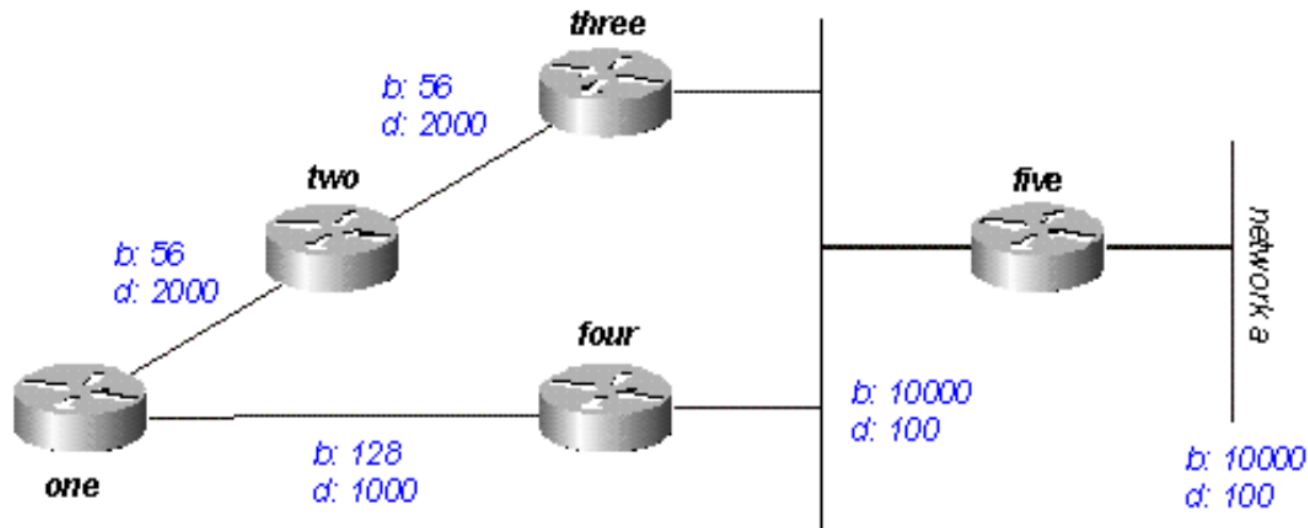


Fig. 2 A sample network

Neighbor Discovery

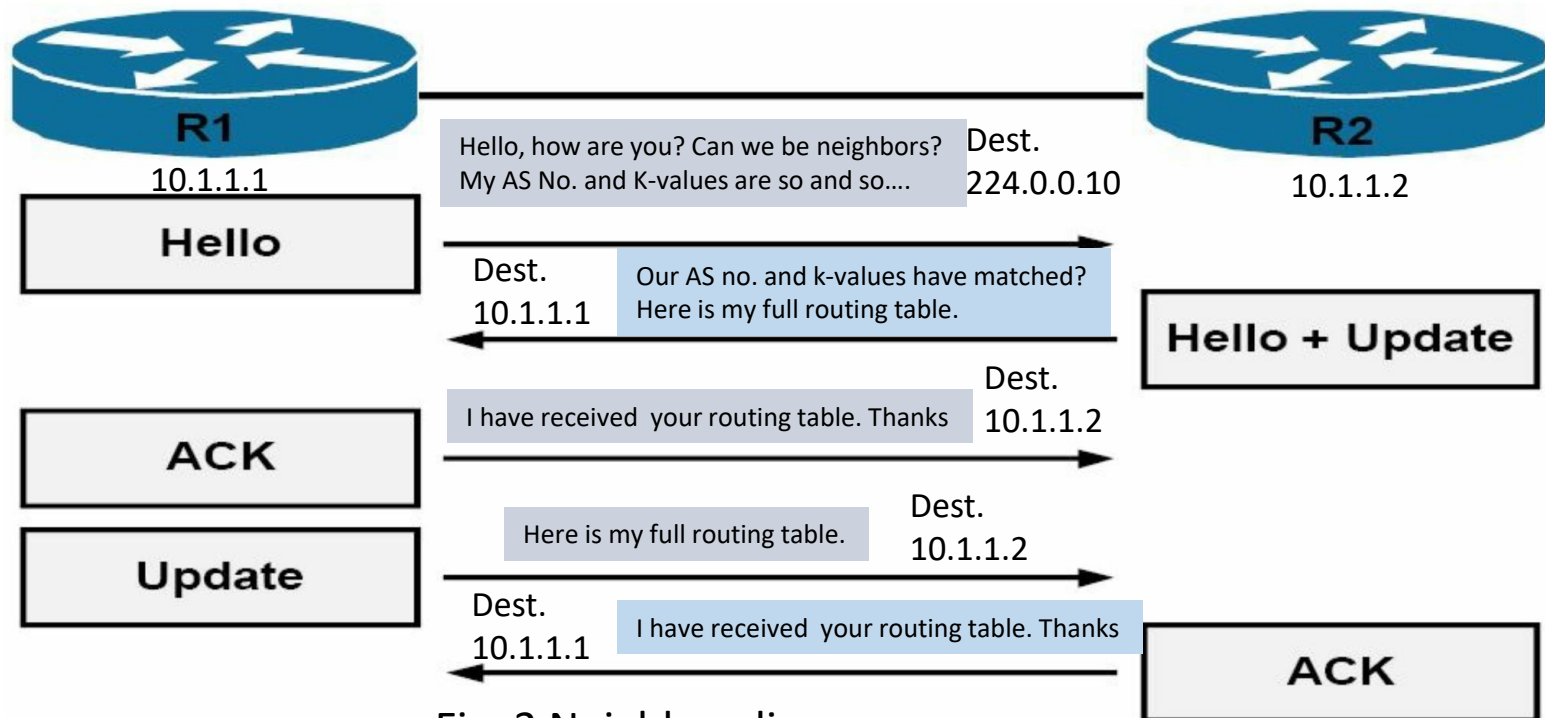


Fig. 3 Neighbor discovery

Neighbor Maintenance



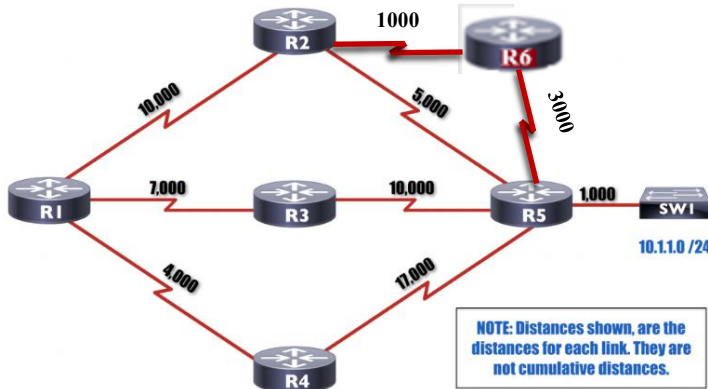
- In **neighbor discovery process**, **full routing table** is sent
- **Otherwise**, only the **change in routing table** is sent
- After a neighbor discovery, the **Hello packet is sent** in every **5 second** to know if the **neighbor is still alive**
- If a router does not receive any Hello packet from a neighbor within **15 seconds (called hold time)**, the **neighbor is considered dead**.
- For low-bandwidth link (e.g., T1), the periods are **60 sec** and **180 sec**.

EIGRP Tables



Feasibility condition: reported distance (RD) must be less than the feasible distance (FD) through the successor

$(RD < FD)$



Neighbor	Reported Distance (RD)	Feasible Distance (FD)	(Feasible) Successor?
R2	6,000	16,000	Yes: Successor
R3	11,000	18,000	Yes: Feasible Successor
R4	18,000	22,000	No

Reported distance (RD):
Distance advertised from neighbor as the distance between the Neighbor and the destination.

Feasible distance (FD):
Sum of Reported distance and distance between the router and the neighbor which reports the distance.

A is the route whose reported distance is less than the feasible distance of the best path.

EIGRP Tables



- Neighbor Table

EIGRP shares **routing information only with neighbors**. To know who the neighbors are, it uses neighbor table. When a new neighbor is discovered, EIGRP would add its address and interface on which neighbor is connected in neighbor table.

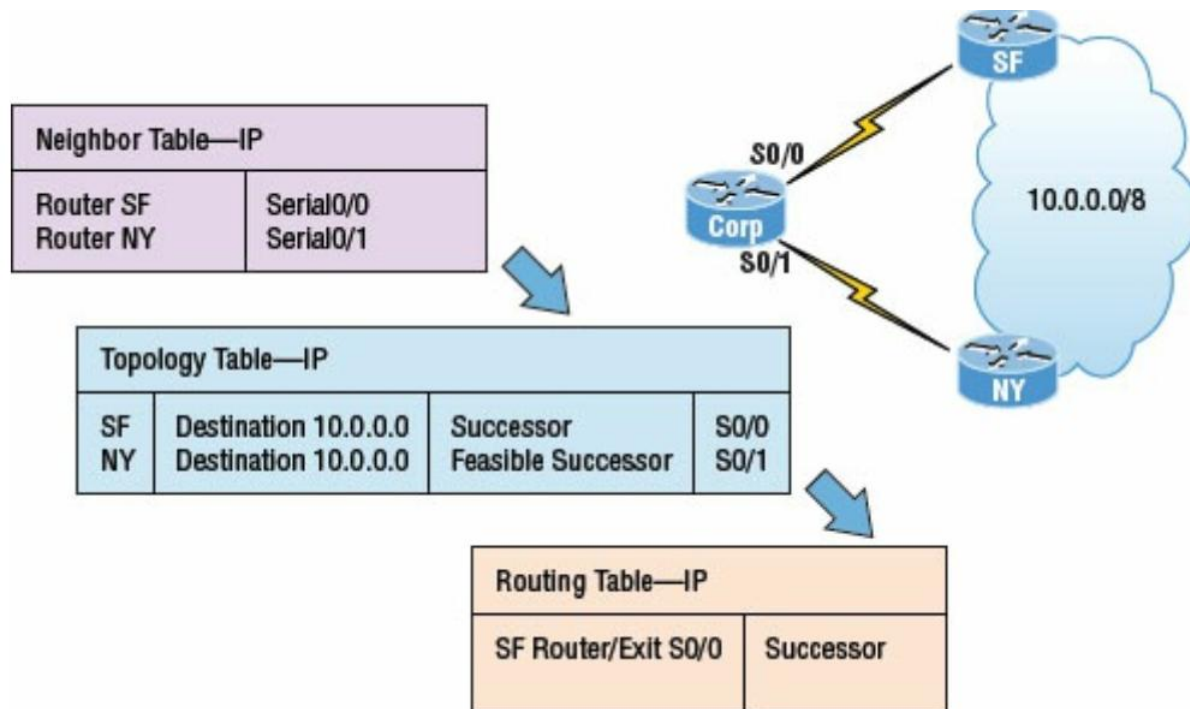
- Topology Table

Stores all **feasible successors** along with the successor (best route) for each destination network. EIGRP can store up to **32 feasible successors**.

- Routing Table

EIGRP **stores single best route** for each destination in this table. Router uses this table to forward the packet.

EIGRP Tables: Example





Routing Protocol: OSPF

EIGRP VS OSPF

Topic sub heading..



EIGRP		OSPF
1	It supports maximum 255 routers in the network. However, the default is 100 routers. (highly scalable)	Supports unlimited number of routers
2	Fast convergence due to feasible successor	Fastest convergence speed due to the area concept
3	Cisco proprietary protocol and can be implemented only in Cisco routers.	Open standard protocol and can be implement in any router.
4	It calculates the metric in terms of bandwidth and delay (default).	It calculates the metric in terms of bandwidth only.
5	EIGRP works on DUAL (Diffusing Update Algorithm) Algorithm.	OSPF works on Dijkstra Algorithm.
6	It maintains the best route and some other alternative routes for each destination.	It maintains the best route in routing table and all routes in database table.
7	It is basically use for medium to lager size organization in the network.	It is basically use for lager size organization in the network.
8	Administrative distance 90	Administrative distance 110
9	Easy to implement	Implementation is complicated

OSPF Area



- An autonomous system (AS) is divided into one or more area.
- Each area is given an **area ID**
- An **AS must have an area having ID 0 (zero)** for multi-area OSPF. Such area is called **backbone area**.
- All areas of an AS must be **connected to the backbone area**.
- A router in an area exchanges routing information with the routers of its area only (by default)

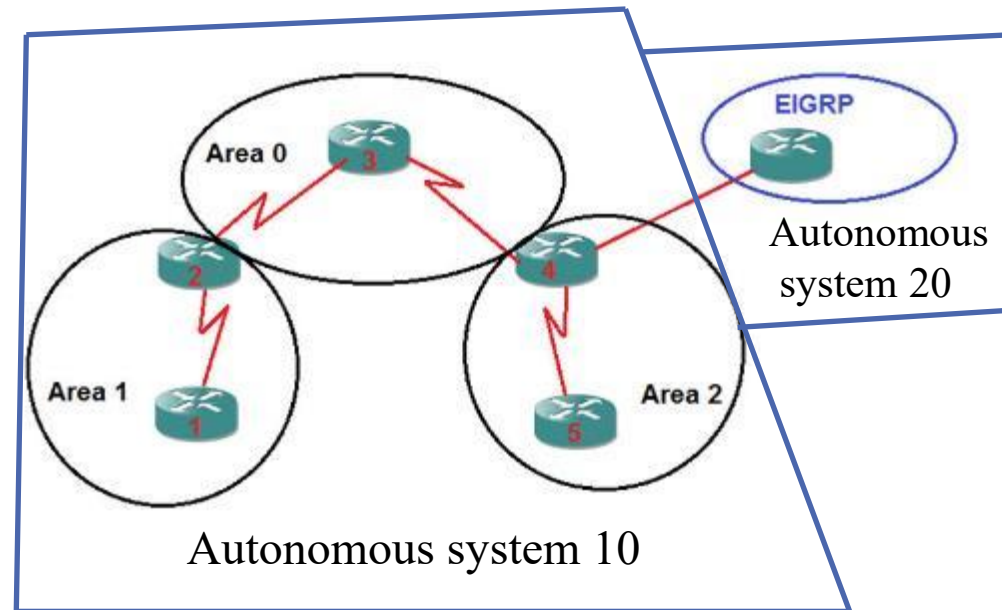
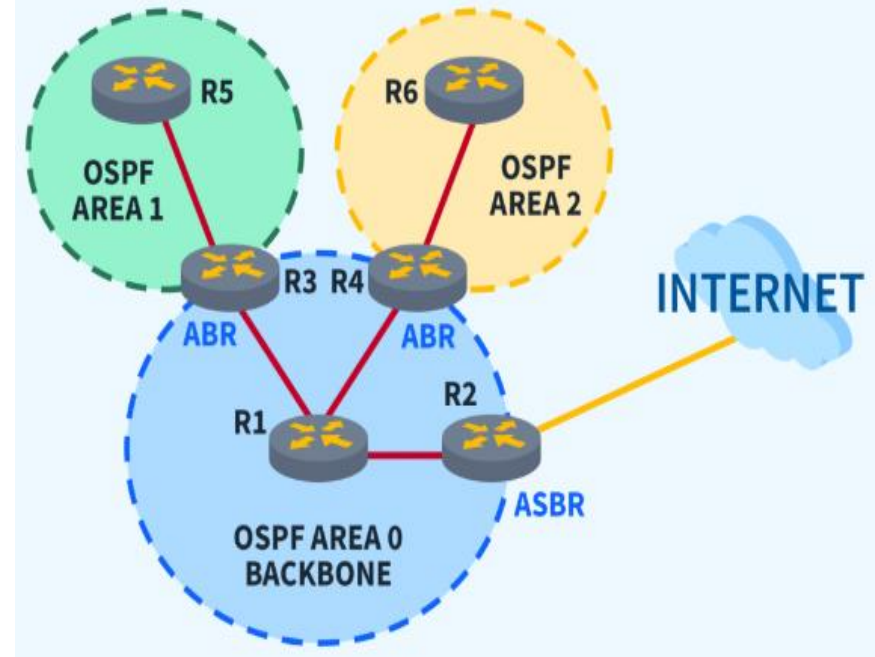


Fig. 1 Autonomous systems and area

OSPF Routers



- **Internal Router (IR):** The router for which all its interface belong to one area. **Router 1 (R1) and Router 5 (R5).**
- **Area Border Router (ABRs):** The router that contains interfaces in more than one area. **Router 2 and Router 4**
- **Backbone Router:** The router that has all or at least one interface in **Area 0**. **Router 3, Router 2 and Router 4.**
- **Autonomous System Boundary Router (ASBR):** The routers with connection to a separate autonomous system. **Router 2** in the example is connected to EIGRP.



OSPF Data Structure & Packets



■ Link state advertisement (LSA)

- A **data structure** with some specific information about the networks.
- Depending on its **type**, it holds information about
 - a router's interfaces,
 - all routers attached to network,
 - summary routing information of an area,
 - all routers of an AS.

■ Link state database (LSDB)

- A collection of all LSAs known to a router
- In a convergent network, all routers of a network have the same LSDB.

Link State Database (LSDB)

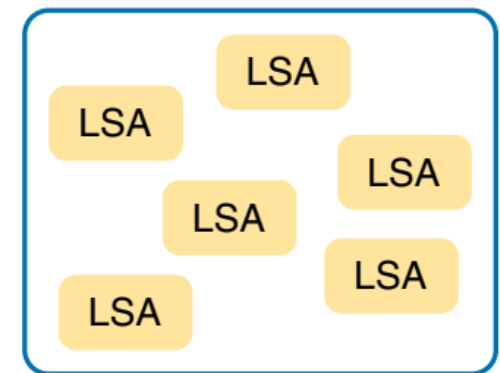


Fig. 2 LSA & LSDB relationship

OSPF Data Structure & Packets



- **Hello**
Used to build and maintain neighbor relationships.
- **DBD – Database Description**
List of LSAs contained in a LSDB. This packet type is circulated when two routers are initially exchanging their link-state databases.
- **Link State Request (LSR)**
Used to request complete information about a link learned from another router.
- **Link State Update (LSU)**
Used to send one or LSA(s)
- **Links State Acknowledgement (LSAck)**
Used to acknowledge the reception of an LSA

Neighbor Discovery



Parameters need to be identical for **two routers to become neighbors**

- **Network mask**—*net mask of the sending router*
- **Subnet number**—derived using the subnet mask and each router's interface Internet Protocol (IP) address
- **Area ID**—*area ID of the sending interface*
- **Hello interval**—*how often Hello packets are transmitted*
- **Dead interval**—*how long to wait for Hello packets before terminating neighbor*
- **Authentication type and password**—*optional*
- **Stub area flag**—*specifies the type of stub area, if applicable*

Hello packet contains all these information



Neighbor Discovery

Example:



The scenario **begins with the link down state**, so the routers have no knowledge of each other as OSPF neighbors

1. Link between R1 and R2 comes up
2. R1 sends the first Hello to multicast IP address **224.0.0.5**, so R2 learns of the existence of R1 as an OSPF router. At that point, R2 lists R1 as a neighbor, with an temporary beginning state of **init state**.

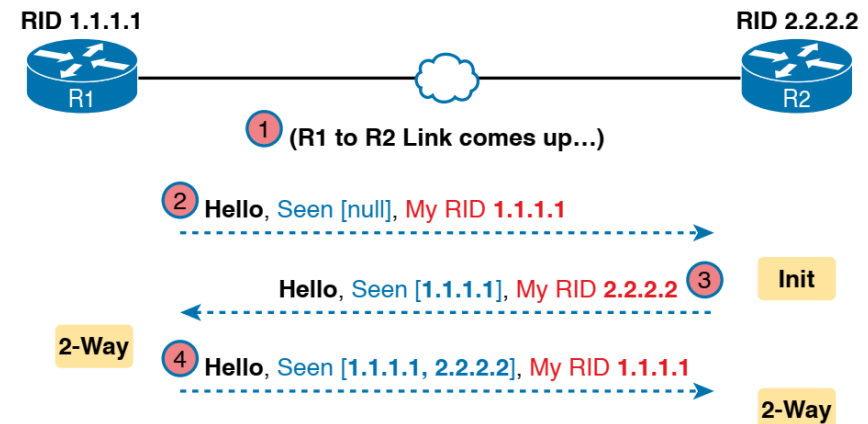


Fig. 3 Neighbor discovery

3. R2 sends back a Hello which tells R1 that R2 exists, and it allows R1 to move through the **init state** and quickly to a **2-way state**.
4. R2 receives the next Hello from R1, and R2 can also move to a 2-way state

Router ID (RID)



- 32-bit unique dotted decimal number

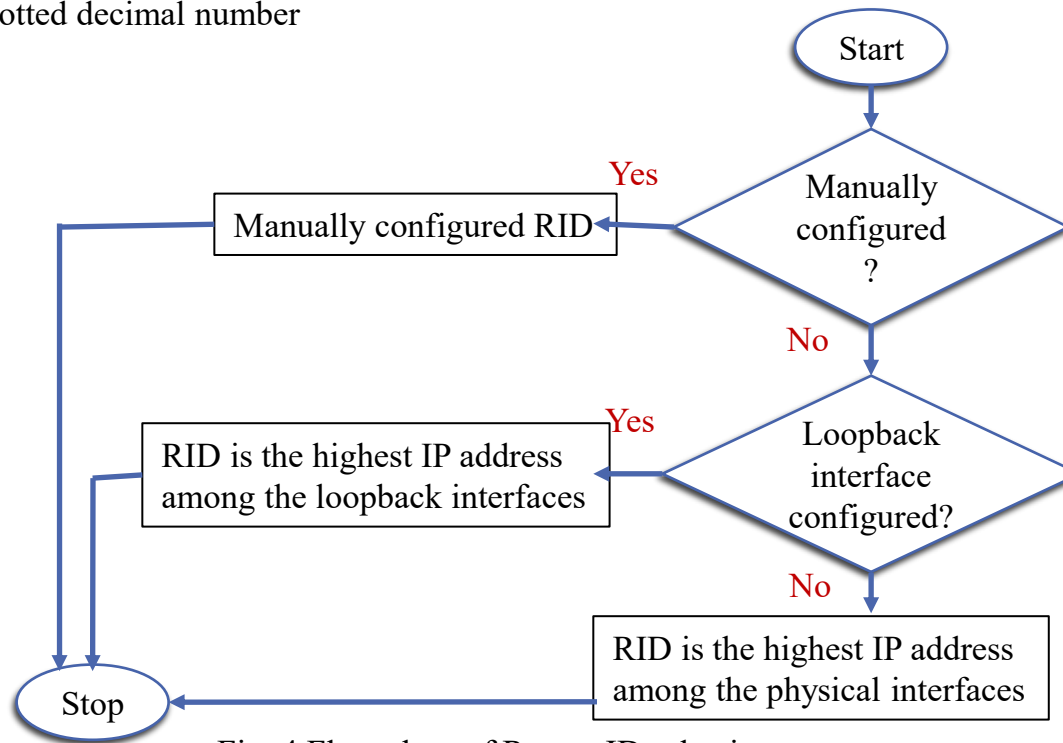


Fig. 4 Flow chart of Router ID selection

DR and BDR

- **Point-to-point network:** A network in **an area** connecting **only two routers directly**.
- **Broadcast network:** A network in **an area** connecting **more than two routers**
- **Designated router (DR):**
 - In a **broadcast network**, a router with the **highest priority**.
 - If the priorities tie, the router has the highest RID (Router ID)
 - All database exchange is done via DR
- **Backup Designated router (BDR)**
 - In a **broadcast network**, a router with the **second highest priority**.
 - If the priorities tie, the router having the second-highest RID
 - If the DR fails, the BDR takes over.
- **DROTHER:** The router which is neither DR nor BDR

DR and BDR

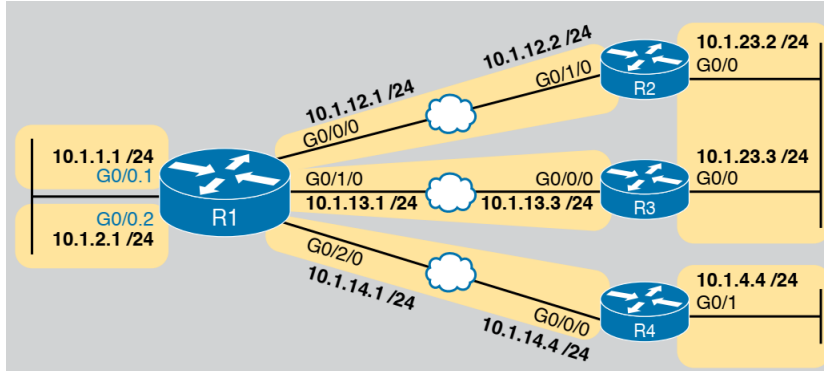


Fig. point-to-point network

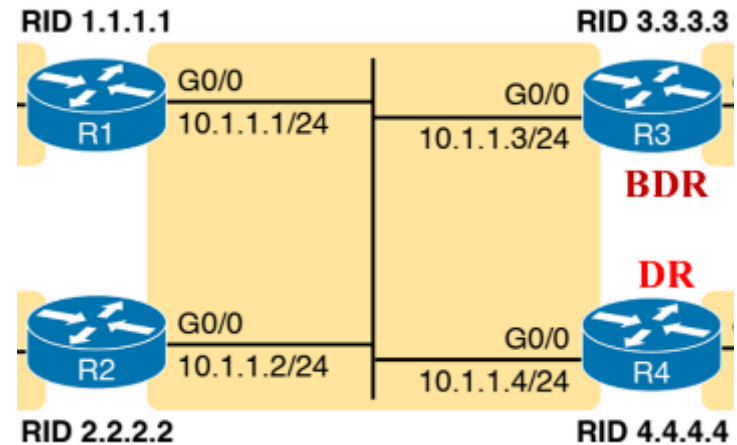


Fig. Broadcast network , DR and BDR election

DR and BDR

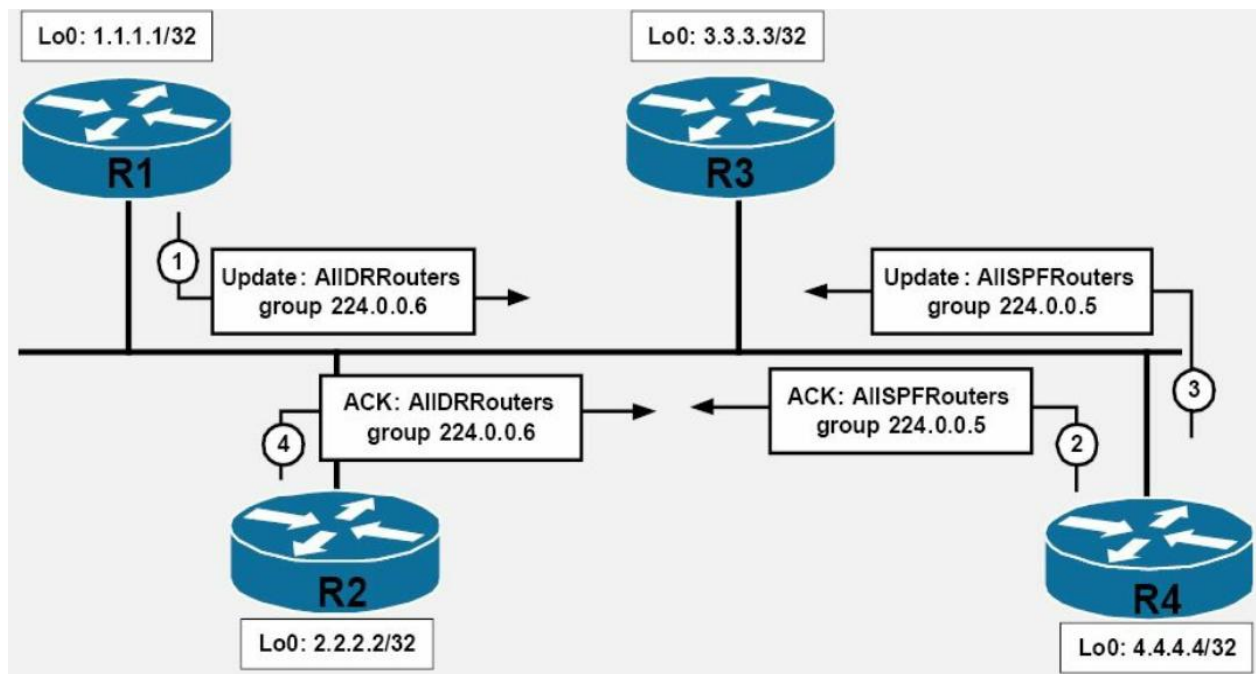


Fig. 7 Illustration of update exchange through DR and BDR

Wildcard Mask



- Used to specify a **range of network addresses**.
- Inverted subnet mask
- Used in EIGRP, OSPF and Access Control Lists (ACLs)
- **How to get wildcard mask of an IP address?**
 Subtract the subnet mask from 255.255.255.255
- **What does each bit of a wildcard mask mean?**
 0 : All IP address in the range must match the bit
 1 : Different IP address in the range can have different value in the bit position

Wildcard Mask



- Only 192.168.3.0

All bits must match.

WCM: All bits 0. (00000000.00000000.00000000.00000000)

WCM: 0.0.0.0

- IP address range: 192.168.3.0 to 192.168.3.255

Match first **three block** (24 bits) and fourth block can take any value

WCM: 0.0.0.255

- IP address range: 192.168.3.4 to 192.168.3.13

11000000.10101000.00000011.00000100

11000000.10101000.00000011.00001101

First **28 bits same**.

Match first 28 bits; make them all zero

Make rest of the bits 1

Actual IP range: 192.168.3.0 to 192.168.3.15 under WCM: 0.0.0.15



References

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- [2] Cisco, “<https://www.cisco.com/c/en/us/support/docs/ip/enhanced-interior...-gateway-routing-protocol-eigrp/16406-eigrp-toc.html>, [Accessed: April. 27, 2020].
- [3] P. Browning, F. Tafa, D. Gheorghe, and D. Barinic, *Cisco CCNA in 60 Days*, Reality Press Ltd., UK, 2014, pp. 581
- [4] Computer Networking Notes, <https://www.computernetworkingnotes.com/ccna...-study-guide/eigrp-tutorial-basic-concept-explained.html>, [Accessed: April. 27, 2020].



Recommended Books

1. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
2. **CCNA Routing and Switching**, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.